

Dimensionality Reduction and Clustering

Controlled Experiments on Synthetic Data and
Validation on Residential Energy Consumption Data



UNIVERSITÀ
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OVERVIEW

Context:

- Unsupervised learning and exploratory data analysis
- Interaction between dimensionality reduction and clustering
- Controlled experiments on synthetic datasets
- Validation on residential energy consumption data

Goal: to evaluate how dimensionality reduction used as pre-processing affects clustering performance and identify robust algorithmic combinations.

DIMENSIONALITY REDUCTION ALGORITHMS

Reducing the number of variables in a dataset

Linear

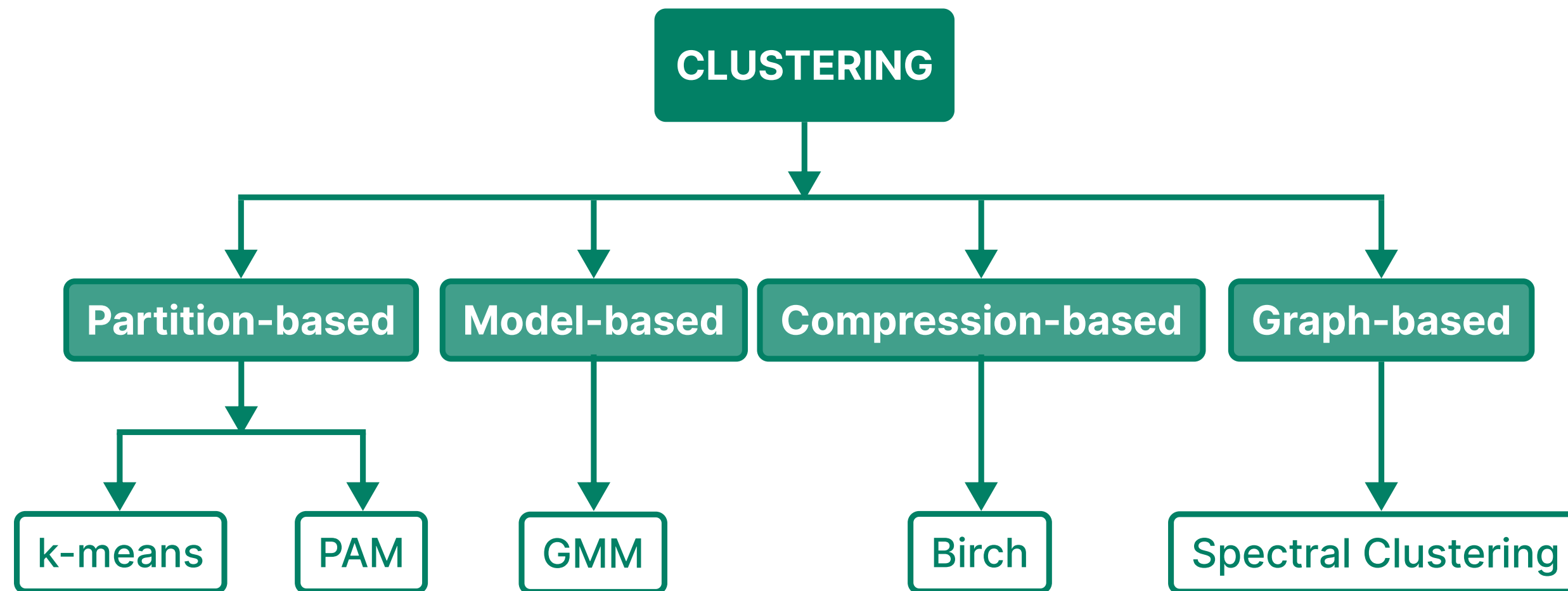
PCA project the data into a lower-dimensional space using linear combinations of the original variables. These methods are computationally efficient and highly interpretable. However, they may fail to capture complex nonlinear relationships among features.

Non linear

t-SNE and **UMAP** are capable of capturing complex, nonlinear relationships by modelling underlying structures of the data. Nonlinear methods are particularly effective for visualization and cluster detection, though they tend to be less immediately interpretable and more sensitive to parameter choices.

CLUSTERING

ALGORITHMS



Goal: group observations into different clusters such that elements within the same group share strong similarities, while those in different groups are clearly dissimilar

VALIDATION MEASURES

To quantitatively evaluate clustering results

External Evaluation - Known Label

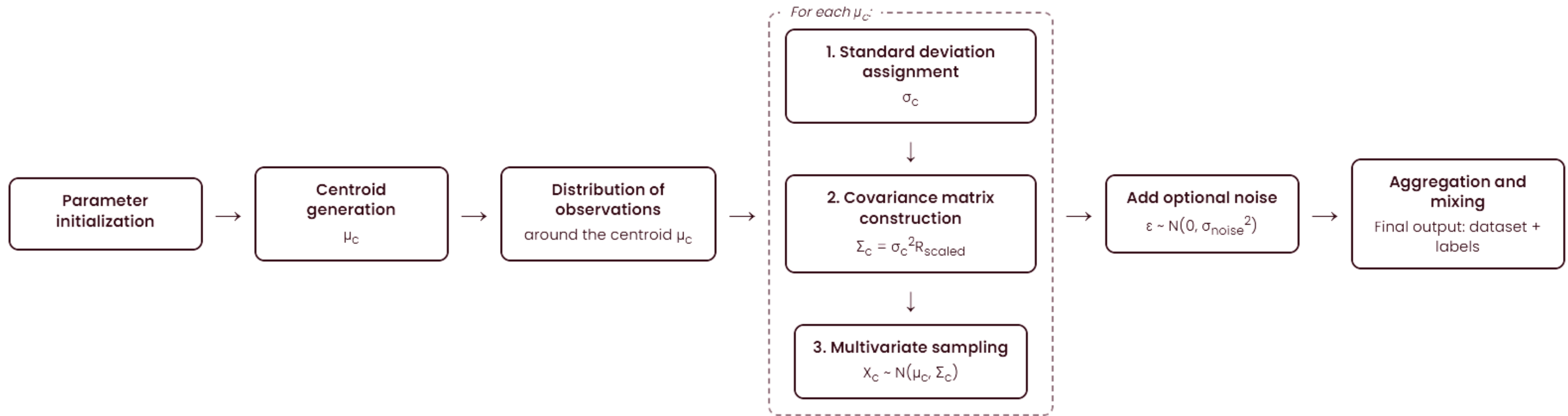
- **Adjusted Rand Index (ARI):** agreement with true labels, corrected for chance (-1 to 1 , higher is better)
- **Normalized Mutual Information (NMI):** shared information between labels and clusters (0 to 1 , higher is better)

Internal Evaluation - Unknown Label

- **Silhouette score:** cohesion versus separation (-1 to 1 , higher is better)
- **Davies-Bouldin Index (DBI):** cluster compactness versus separation (≥ 0 , lower is better)
- **Calinski-Harabasz Index (CHI):** between versus within variance (≥ 0 , higher is better)

1 Controlled Experiments on Synthetic Data

Application on Controlled Settings



- Random centroids uniformly distributed in feature space
- Minimum separation constraint to ensure cluster separability
- Multivariate Gaussian sampling per cluster
- Positive definite covariance matrix
- Controlled variance and feature correlation
- Flexible sample distribution across clusters
- Optional zero-mean isotropic Gaussian noise

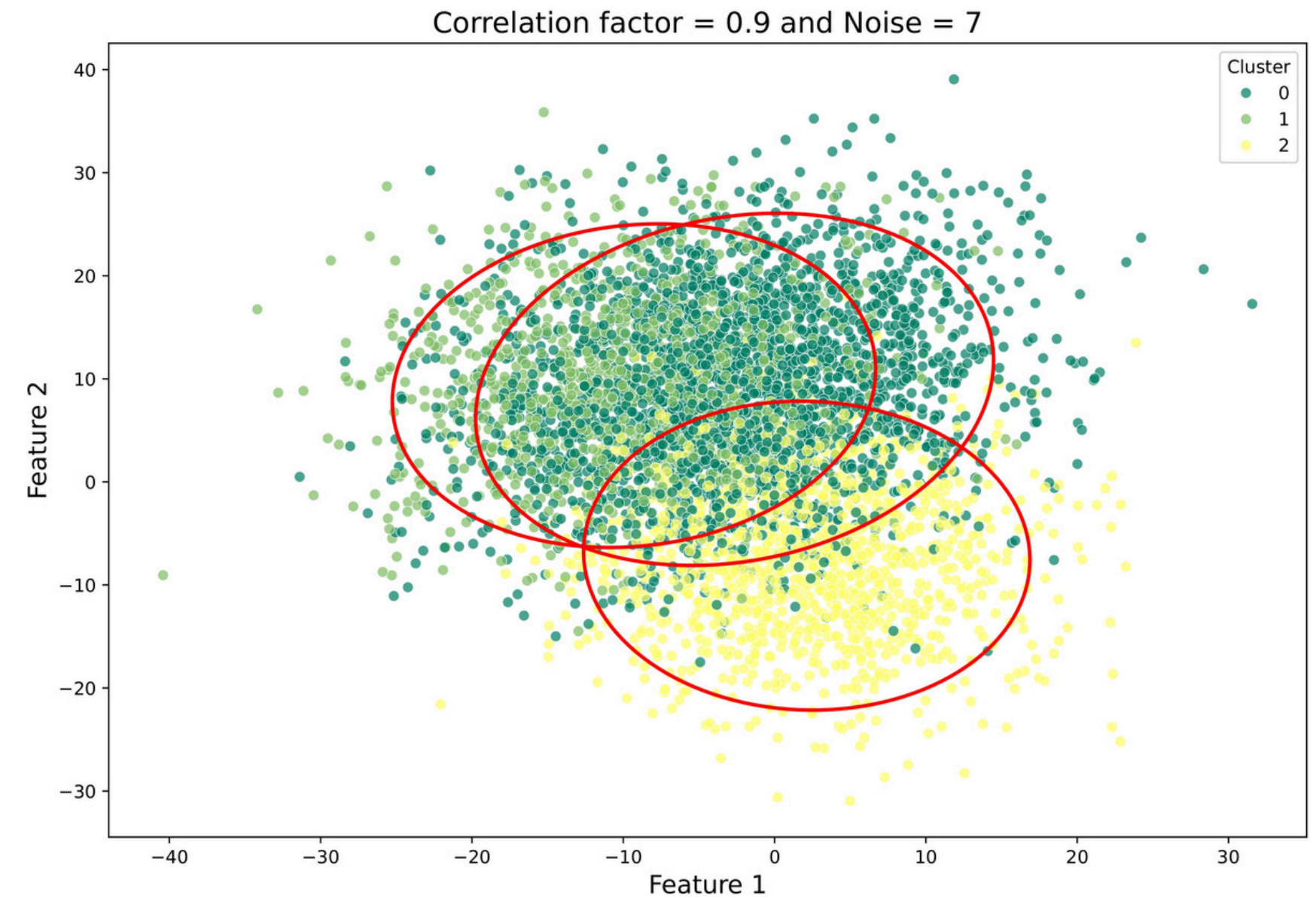
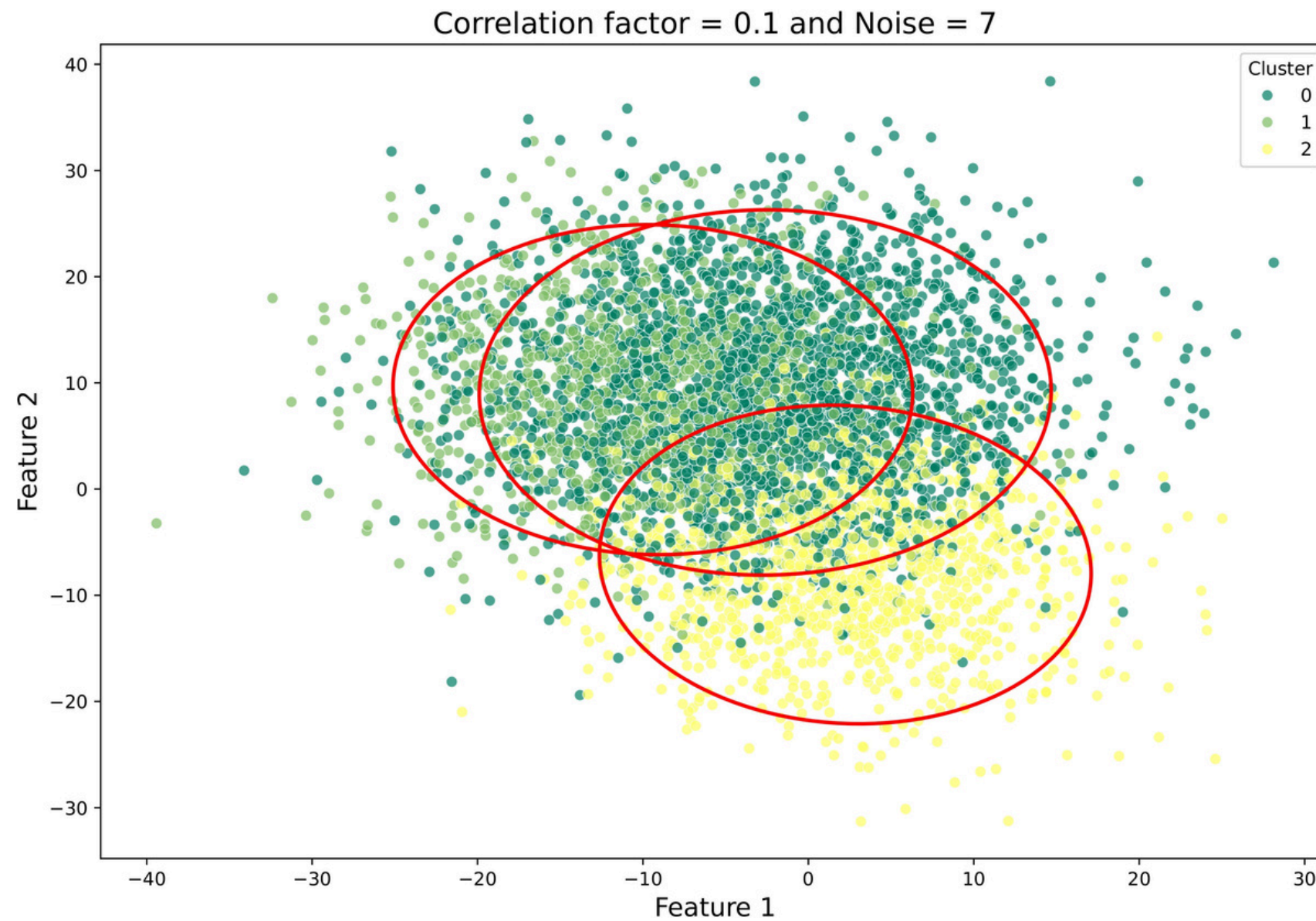
PIPELINE

SYNTHETIC DATA

k	2500 × 15	5000 × 10	10000 × 5
3	1250 $\sigma = 5$ 625 $\sigma = 3.5$ 625 $\sigma = 2$	2500 $\sigma = 5$ 1250 $\sigma = 3.5$ 1250 $\sigma = 2$	5000 $\sigma = 5$ 2500 $\sigma = 3.5$ 2500 $\sigma = 2$
5	1000 $\sigma = 5$ 500 $\sigma = 3.5$ 500 $\sigma = 2$ 2500 $\sigma = 3.5$ 2500 $\sigma = 2$	2000 $\sigma = 5$ 1000 $\sigma = 3.5$ 1000 $\sigma = 2$ 500 $\sigma = 3.5$ 500 $\sigma = 2$	4000 $\sigma = 5$ 2000 $\sigma = 3.5$ 2000 $\sigma = 2$ 1000 $\sigma = 3.5$ 1000 $\sigma = 2$
7	900 $\sigma = 5$ 450 $\sigma = 3.5$ 450 $\sigma = 2$ 225 $\sigma = 3.5$ 225 $\sigma = 2$ 125 $\sigma = 3.5$ 125 $\sigma = 2$	1800 $\sigma = 5$ 900 $\sigma = 3.5$ 900 $\sigma = 2$ 450 $\sigma = 3.5$ 450 $\sigma = 2$ 250 $\sigma = 3.5$ 250 $\sigma = 2$	3600 $\sigma = 5$ 1800 $\sigma = 3.5$ 1800 $\sigma = 2$ 900 $\sigma = 3.5$ 900 $\sigma = 2$ 500 $\sigma = 3.5$ 500 $\sigma = 2$

- One dominant cluster (higher cardinality and greater variance)
- Two equally sized minor clusters with different within-cluster dispersion
- Same structural pattern repeated regardless of the number of clusters and dataset dimensionality

CORRELATION FACTOR



RESULTS

Performance for 2.500 observations and 15 variables, evaluated using ARI, keeping noise constant at 7.

- Overall positive results, with few isolated underperforming cases
- Dimensionality reduction has a strong impact on PAM
- A higher correlation factor leads to lower ARI (for a fixed number of clusters)
- A larger number of clusters results in performance deterioration
- UMAP emerges as the most effective preprocessing method, stabilizing and often improving clustering performance

Observations and Features:	2500 observations and 15 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Correlation Factor:	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9
k-means	0,890	0,897	0,414	0,570	0,795	0,718	0,814	0,626	0,605
PCA + k-means	0,882	0,413	0,876	0,566	0,551	0,656	0,606	0,587	0,551
t-SNE + k-means	0,811	0,802	0,760	0,696	0,517	0,567	0,548	0,468	0,491
UMAP + k-means	0,901	0,892	0,906	0,577	0,837	0,571	0,759	0,591	0,611
Birch	0,819	0,843	0,835	0,684	0,520	0,533	0,740	0,537	0,534
PCA + Birch	0,852	0,883	0,895	0,808	0,530	0,513	0,720	0,543	0,492
t-SNE + Birch	0,808	0,822	0,743	0,632	0,578	0,496	0,613	0,393	0,511
UMAP + Birch	0,901	0,892	0,906	0,788	0,662	0,674	0,594	0,699	0,600
GMM	0,914	0,928	0,448	0,750	0,679	0,835	0,515	0,504	0,619
PCA + GMM	0,900	0,901	0,914	0,704	0,480	0,787	0,520	0,605	0,551
t-SNE + GMM	0,384	0,820	0,871	0,639	0,557	0,576	0,426	0,472	0,486
UMAP + GMM	0,895	0,899	0,918	0,603	0,835	0,572	0,700	0,629	0,571
PAM	0,563	0,427	0,708	0,485	0,473	0,458	0,418	0,421	0,358
PCA + PAM	0,884	0,889	0,879	0,810	0,533	0,545	0,606	0,591	0,549
t-SNE + PAM	0,816	0,808	0,782	0,554	0,516	0,567	0,556	0,475	0,571
UMAP + PAM	0,900	0,894	0,905	0,835	0,559	0,848	0,609	0,613	0,621
Spectral Clustering	0,909	0,915	0,907	0,846	0,853	0,872	0,827	0,825	0,628
PCA + Spectral Clustering	0,896	0,909	0,906	0,835	0,806	0,563	0,618	0,593	0,564
t-SNE + Spectral Clustering	0,727	0,763	0,590	0,485	0,475	0,561	0,480	0,443	0,579
UMAP + Spectral Clustering	0,890	0,898	0,910	0,837	0,836	0,854	0,813	0,623	0,582

RESULTS

Performance for 5.000 observations and 10 variables, evaluated using ARI, keeping noise constant at 7.

- Increasing observations and reducing variables leads to overall performance decline
- t-SNE appears more unstable compared to other preprocessing methods
- UMAP more frequently stabilizes or improves results
- Spectral Clustering shows the best individual performance

Observations and Features:	5000 observations and 10 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Correlation Factor:	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9
k-means	0,512	0,472	0,418	0,363	0,344	0,343	0,380	0,364	0,324
PCA + k-means	0,520	0,456	0,415	0,343	0,278	0,261	0,257	0,269	0,240
t-SNE + k-means	0,362	0,434	0,382	0,316	0,322	0,305	0,263	0,254	0,244
UMAP + k-means	0,488	0,446	0,399	0,453	0,342	0,434	0,408	0,307	0,293
Birch	0,459	0,433	0,405	0,336	0,291	0,291	0,202	0,239	0,232
PCA + Birch	0,372	0,306	0,399	0,308	0,276	0,267	0,232	0,242	0,192
t-SNE + Birch	0,440	0,304	0,354	0,304	0,361	0,288	0,276	0,207	0,234
UMAP + Birch	0,490	0,439	0,356	0,393	0,289	0,270	0,380	0,388	0,303
GMM	0,380	0,416	0,421	0,360	0,415	0,318	0,261	0,267	0,203
PCA + GMM	0,519	0,345	0,436	0,260	0,283	0,237	0,220	0,261	0,203
t-SNE + GMM	0,286	0,377	0,331	0,294	0,291	0,318	0,254	0,263	0,271
UMAP + GMM	0,456	0,371	0,343	0,302	0,305	0,390	0,364	0,259	0,242
PAM	0,463	0,393	0,343	0,290	0,305	0,285	0,261	0,289	0,243
PCA + PAM	0,522	0,457	0,414	0,353	0,261	0,260	0,258	0,233	0,231
t-SNE + PAM	0,423	0,433	0,390	0,314	0,324	0,295	0,271	0,235	0,240
UMAP + PAM	0,496	0,438	0,403	0,463	0,461	0,437	0,406	0,324	0,300
Spectral Clustering	0,552	0,446	0,390	0,492	0,449	0,415	0,492	0,444	0,406
PCA + Spectral Clustering	0,505	0,448	0,419	0,333	0,273	0,216	0,250	0,215	0,191
t-SNE + Spectral Clustering	0,319	0,371	0,380	0,294	0,279	0,292	0,252	0,221	0,237
UMAP + Spectral Clustering	0,494	0,397	0,381	0,434	0,438	0,431	0,471	0,442	0,385

RESULTS

Performance for 10.000 observations and 5 variables, evaluated using ARI, keeping noise constant at 7.

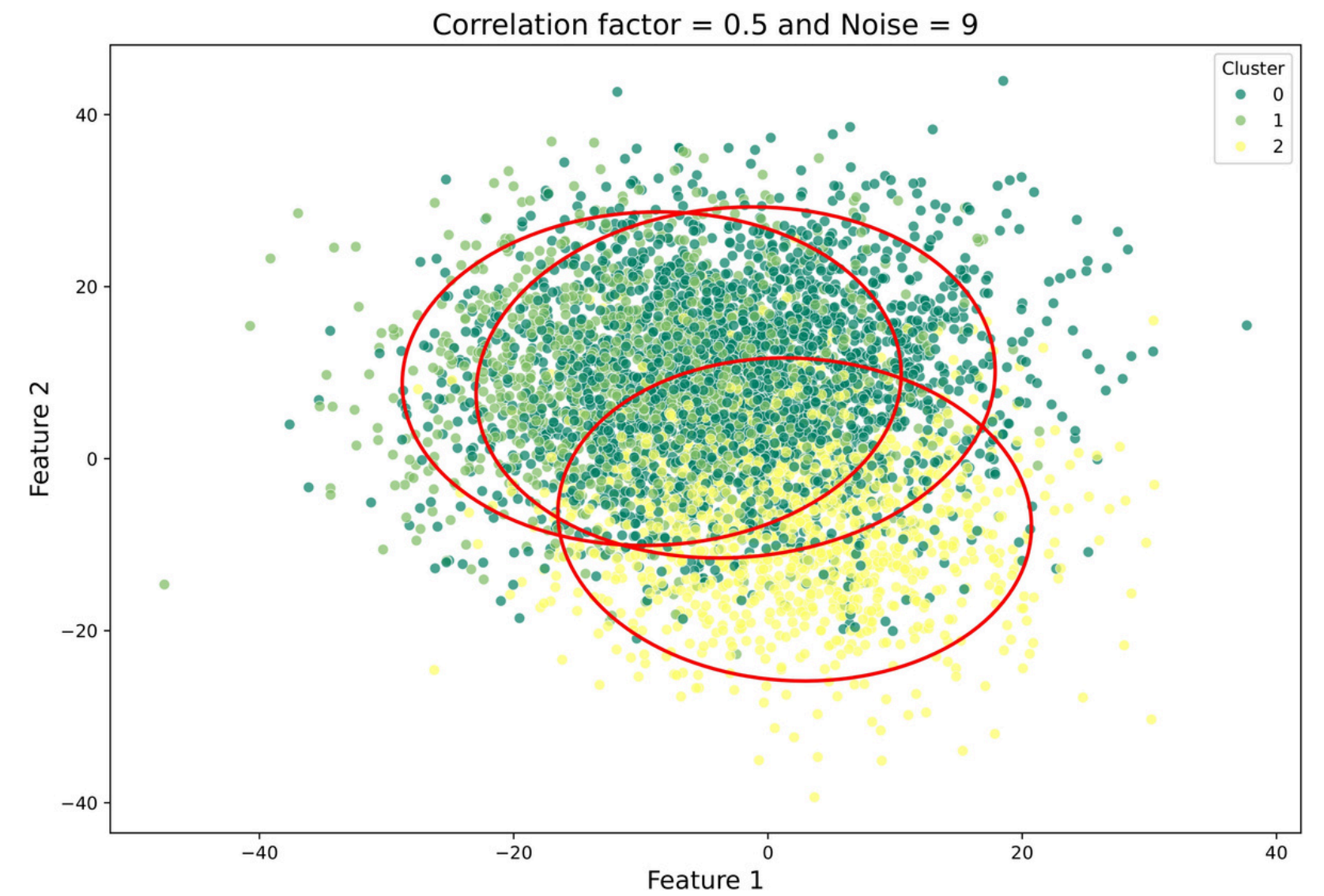
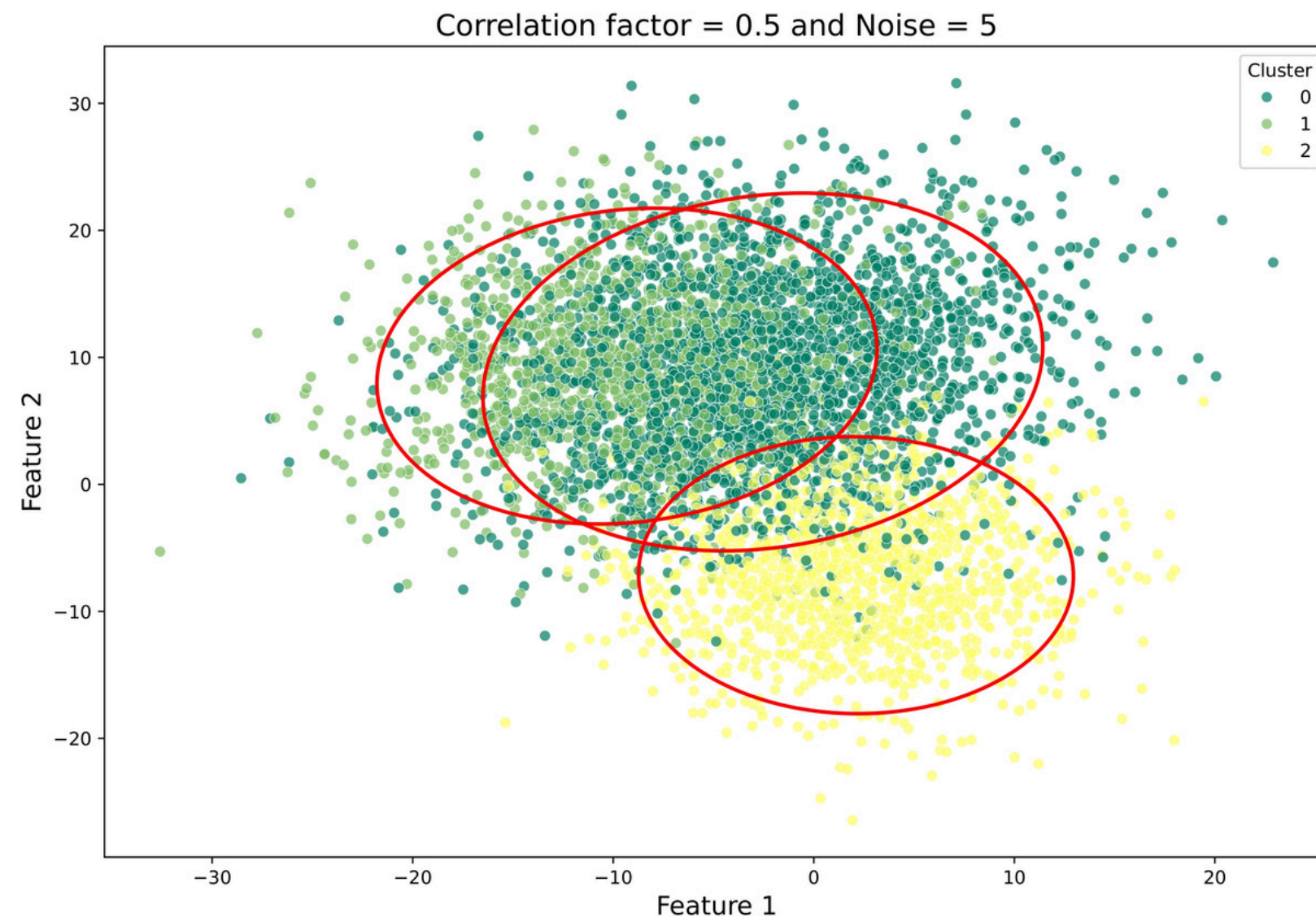
- More observations combined with fewer informative variables leads to worse results
- t-SNE again shows unfavorable behavior
- Reduced impact of dimensionality reduction methods overall
- Internal feature correlation negatively affects performance
- With fewer informative variables, correlation reduces effective dimensionality and weakens cluster separation

Observations and Features:	10000 observations and 5 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Correlation Factor:	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9
k-means	0,319	0,299	0,260	0,206	0,209	0,216	0,190	0,186	0,198
PCA + k-means	0,307	0,272	0,254	0,213	0,208	0,157	0,150	0,145	0,169
t-SNE + k-means	0,273	0,251	0,229	0,205	0,189	0,195	0,186	0,178	0,162
UMAP + k-means	0,304	0,304	0,267	0,223	0,226	0,204	0,187	0,204	0,183
Birch	0,303	0,287	0,225	0,165	0,132	0,160	0,137	0,169	0,108
PCA + Birch	0,289	0,255	0,242	0,171	0,186	0,120	0,215	0,204	0,172
t-SNE + Birch	0,198	0,078	0,270	0,141	0,191	0,119	0,164	0,169	0,148
UMAP + Birch	0,290	0,252	0,266	0,231	0,192	0,204	0,175	0,225	0,188
GMM	0,233	0,264	0,221	0,141	0,187	0,139	0,171	0,082	0,150
PCA + GMM	0,049	0,214	0,209	0,173	0,148	0,144	0,195	0,207	0,155
t-SNE + GMM	0,112	0,222	0,073	0,207	0,153	0,162	0,138	0,185	0,158
UMAP + GMM	0,310	0,270	0,400	0,249	0,193	0,130	0,165	0,196	0,184
PAM	0,318	0,349	0,250	0,228	0,200	0,230	0,190	0,188	0,184
PCA + PAM	0,312	0,341	0,271	0,213	0,204	0,164	0,162	0,153	0,168
t-SNE + PAM	0,279	0,251	0,233	0,208	0,192	0,151	0,186	0,180	0,162
UMAP + PAM	0,307	0,300	0,269	0,221	0,226	0,205	0,187	0,211	0,185
Spectral Clustering	0,334	0,310	0,257	0,218	0,204	0,216	0,184	0,194	0,199
PCA + Spectral Clustering	0,311	0,293	0,258	0,208	0,196	0,153	0,153	0,142	0,164
t-SNE + Spectral Clustering	0,203	0,000	0,085	0,176	0,167	0,184	0,152	0,195	0,150
UMAP + Spectral Clustering	0,324	0,313	0,272	0,221	0,232	0,207	0,166	0,196	0,193

RESULTS

Observations and Features:	2500 observations and 15 features									5000 observations and 10 features									10000 observations and 5 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters			3 clusters			5 clusters			7 clusters			3 clusters			5 clusters			7 clusters		
Correlation Factor:	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9	0,1	0,5	0,9
k-means	0,890	0,897	0,414	0,570	0,795	0,718	0,814	0,626	0,605	0,512	0,472	0,418	0,363	0,344	0,343	0,380	0,364	0,324	0,319	0,299	0,260	0,206	0,209	0,216	0,190	0,186	0,198
PCA + k-means	0,882	0,413	0,876	0,566	0,551	0,656	0,606	0,587	0,551	0,520	0,456	0,415	0,343	0,278	0,261	0,257	0,269	0,240	0,307	0,272	0,254	0,213	0,208	0,157	0,150	0,145	0,169
t-SNE + k-means	0,811	0,802	0,760	0,696	0,517	0,567	0,548	0,468	0,491	0,362	0,434	0,382	0,316	0,322	0,305	0,263	0,254	0,244	0,273	0,251	0,229	0,205	0,189	0,195	0,186	0,178	0,162
UMAP + k-means	0,901	0,892	0,906	0,577	0,837	0,571	0,759	0,591	0,611	0,488	0,446	0,399	0,453	0,342	0,434	0,408	0,307	0,293	0,304	0,304	0,267	0,223	0,226	0,204	0,187	0,204	0,183
Birch	0,819	0,843	0,835	0,684	0,520	0,533	0,740	0,537	0,534	0,459	0,433	0,405	0,336	0,291	0,291	0,202	0,239	0,232	0,303	0,287	0,225	0,165	0,132	0,160	0,137	0,169	0,108
PCA + Birch	0,852	0,883	0,895	0,808	0,530	0,513	0,720	0,543	0,492	0,372	0,306	0,399	0,308	0,276	0,267	0,232	0,242	0,192	0,289	0,255	0,242	0,171	0,186	0,120	0,215	0,204	0,172
t-SNE + Birch	0,808	0,822	0,743	0,632	0,578	0,496	0,613	0,393	0,511	0,440	0,304	0,354	0,304	0,361	0,288	0,276	0,207	0,234	0,198	0,078	0,270	0,141	0,191	0,119	0,164	0,169	0,148
UMAP + Birch	0,901	0,892	0,906	0,788	0,662	0,674	0,594	0,699	0,600	0,490	0,439	0,356	0,393	0,289	0,270	0,380	0,388	0,303	0,290	0,252	0,266	0,231	0,192	0,204	0,175	0,225	0,188
GMM	0,914	0,928	0,448	0,750	0,679	0,835	0,515	0,504	0,619	0,380	0,416	0,421	0,360	0,415	0,318	0,261	0,267	0,203	0,233	0,264	0,221	0,141	0,187	0,139	0,171	0,082	0,150
PCA + GMM	0,900	0,901	0,914	0,704	0,480	0,787	0,520	0,605	0,551	0,519	0,345	0,436	0,260	0,283	0,237	0,220	0,261	0,203	0,049	0,214	0,209	0,173	0,148	0,144	0,195	0,207	0,155
t-SNE + GMM	0,384	0,820	0,871	0,639	0,557	0,576	0,426	0,472	0,486	0,286	0,377	0,331	0,294	0,291	0,318	0,254	0,263	0,271	0,112	0,222	0,073	0,207	0,153	0,162	0,138	0,185	0,158
UMAP + GMM	0,895	0,899	0,918	0,603	0,835	0,572	0,700	0,629	0,571	0,456	0,371	0,343	0,302	0,305	0,390	0,364	0,259	0,242	0,310	0,270	0,400	0,249	0,193	0,130	0,165	0,196	0,184
PAM	0,563	0,427	0,708	0,485	0,473	0,458	0,418	0,421	0,358	0,463	0,393	0,343	0,290	0,305	0,285	0,261	0,289	0,243	0,318	0,349	0,250	0,228	0,200	0,230	0,190	0,188	0,184
PCA + PAM	0,884	0,889	0,879	0,810	0,533	0,545	0,606	0,591	0,549	0,522	0,457	0,414	0,353	0,261	0,260	0,258	0,233	0,231	0,312	0,341	0,271	0,213	0,204	0,164	0,162	0,153	0,168
t-SNE + PAM	0,816	0,808	0,782	0,554	0,516	0,567	0,556	0,475	0,571	0,423	0,433	0,390	0,314	0,324	0,295	0,271	0,235	0,240	0,279	0,251	0,233	0,208	0,192	0,151	0,186	0,180	0,162
UMAP + PAM	0,900	0,894	0,905	0,835	0,559	0,848	0,609	0,613	0,621	0,496	0,438	0,403	0,463	0,461	0,437	0,406	0,324	0,300	0,307	0,300	0,269	0,221	0,226	0,205	0,187	0,211	0,185
Spectral Clustering	0,909	0,915	0,907	0,846	0,853	0,872	0,827	0,825	0,628	0,552	0,446	0,390	0,492	0,449	0,415	0,492	0,444	0,406	0,334	0,310	0,257	0,218	0,204	0,216	0,184	0,194	0,199
PCA + Spectral Clustering	0,896	0,909	0,906	0,835	0,806	0,563	0,618	0,593	0,564	0,505	0,448	0,419	0,333	0,273	0,216	0,250	0,215	0,191	0,311	0,293	0,258	0,208	0,196	0,153	0,153	0,142	0,164
t-SNE + Spectral Clustering	0,727	0,763	0,590	0,485	0,475	0,561	0,480	0,443	0,579	0,319	0,371	0,380	0,294	0,279	0,292	0,252	0,221	0,237	0,203	0,000	0,085	0,176	0,167	0,184	0,152	0,195	0,150
UMAP + Spectral Clustering	0,890	0,898	0,910	0,837	0,836	0,854	0,813	0,623	0,582	0,494	0,397	0,381	0,434	0,438	0,431	0,471	0,442	0,385	0,324	0,313	0,272	0,221	0,232	0,207	0,166	0,196	0,193

Clustering performance measured by the Adjusted Rand Index (ARI)



NOISE

RESULTS

Performance for 2.500 observations and 15 variables, evaluated using ARI, keeping correlation factor at 0.5.

- PAM benefits most from dimensionality reduction
- Increasing noise leads to performance decline across all algorithms
- Noise impact is stronger as the number of clusters increases
- UMAP remains the most stable preprocessing method
- Spectral Clustering shows greater robustness at intermediate noise levels

Observations and Features:	2500 observations and 15 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Noise:	5	7	9	5	7	9	5	7	9
k-means	0,962	0,897	0,785	0,646	0,795	0,475	0,723	0,626	0,492
PCA + k-means	0,96	0,413	0,778	0,881	0,551	0,466	0,693	0,587	0,465
t-SNE + k-means	0,95	0,802	0,637	0,643	0,517	0,416	0,715	0,468	0,327
UMAP + k-means	0,961	0,892	0,792	0,949	0,837	0,466	0,73	0,591	0,372
Birch	0,925	0,843	0,676	0,92	0,52	0,561	0,794	0,537	0,4
PCA + Birch	0,963	0,883	0,731	0,637	0,53	0,397	0,658	0,543	0,268
t-SNE + Birch	0,94	0,822	0,67	0,658	0,578	0,418	0,706	0,393	0,315
UMAP + Birch	0,961	0,892	0,791	0,862	0,662	0,551	0,862	0,699	0,554
GMM	0,976	0,928	0,441	0,655	0,679	0,521	0,656	0,504	0,267
PCA + GMM	0,972	0,901	0,56	0,92	0,48	0,473	0,633	0,605	0,444
t-SNE + GMM	0,959	0,82	0,478	0,699	0,557	0,411	0,886	0,472	0,341
UMAP + GMM	0,961	0,899	0,792	0,953	0,835	0,493	0,735	0,629	0,293
PAM	0,887	0,427	0,4	0,899	0,473	0,24	0,647	0,421	0,222
PCA + PAM	0,961	0,889	0,813	0,613	0,533	0,446	0,688	0,591	0,459
t-SNE + PAM	0,95	0,808	0,662	0,643	0,516	0,424	0,717	0,475	0,321
UMAP + PAM	0,961	0,894	0,793	0,95	0,559	0,609	0,729	0,613	0,482
Spectral Clustering	0,971	0,915	0,794	0,957	0,853	0,685	0,949	0,825	0,674
PCA + Spectral Clustering	0,972	0,909	0,789	0,92	0,806	0,62	0,705	0,593	0,457
t-SNE + Spectral Clustering	0,961	0,763	0,582	0,938	0,475	0,33	0,946	0,443	0,287
UMAP + Spectral Clustering	0,445	0,898	0,796	0,954	0,836	0,565	0,703	0,623	0,444

RESULTS

Performance for 5000 observations and 10 variables, evaluated using ARI, keeping correlation factor at 0.5.

- Overall performance lower than in the 15-variable scenario
- Dimensionality reduction less impactful than previously observed
- UMAP remains slightly more stable, especially under high noise
- Increasing noise leads to a near-monotonic ARI decline
- High noise combined with more clusters amplifies performance deterioration

Observations and Features:	5000 observations and 10 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Noise:	5	7	9	5	7	9	5	7	9
k-means	0,547	0,472	0,378	0,575	0,344	0,284	0,544	0,364	0,239
PCA + k-means	0,533	0,456	0,367	0,389	0,278	0,267	0,36	0,269	0,182
t-SNE + k-means	0,536	0,434	0,28	0,522	0,322	0,253	0,458	0,254	0,177
UMAP + k-means	0,549	0,446	0,355	0,622	0,342	0,334	0,473	0,307	0,198
Birch	0,618	0,433	0,244	0,401	0,291	0,261	0,447	0,239	0,214
PCA + Birch	0,397	0,306	0,285	0,352	0,276	0,24	0,331	0,242	0,169
t-SNE + Birch	0,52	0,304	0,222	0,403	0,361	0,157	0,359	0,207	0,179
UMAP + Birch	0,549	0,439	0,345	0,449	0,289	0,301	0,527	0,388	0,237
GMM	0,683	0,416	0,287	0,49	0,415	0,341	0,528	0,267	0,166
PCA + GMM	0,491	0,345	0,209	0,392	0,283	0,183	0,39	0,261	0,194
t-SNE + GMM	0,44	0,377	0,253	0,536	0,291	0,238	0,421	0,263	0,157
UMAP + GMM	0,494	0,371	0,251	0,511	0,305	0,231	0,47	0,259	0,237
PAM	0,396	0,393	0,298	0,389	0,305	0,225	0,479	0,289	0,177
PCA + PAM	0,544	0,457	0,368	0,328	0,261	0,261	0,345	0,233	0,17
t-SNE + PAM	0,537	0,433	0,291	0,399	0,324	0,252	0,45	0,235	0,18
UMAP + PAM	0,573	0,438	0,337	0,613	0,461	0,263	0,584	0,324	0,203
Spectral Clustering	0,56	0,446	0,354	0,61	0,449	0,317	0,647	0,444	0,247
PCA + Spectral Clustering	0,54	0,448	0,367	0,325	0,273	0,248	0,327	0,215	0,161
t-SNE + Spectral Clustering	0,547	0,371	0,33	0,348	0,279	0,217	0,454	0,221	0,152
UMAP + Spectral Clustering	0,558	0,397	0,323	0,632	0,438	0,278	0,473	0,442	0,233

RESULTS

Performance for 5000 observations and 10 variables, evaluated using ARI, keeping correlation factor at 0.5.

- High overall difficulty, especially with high noise and many clusters
- ARI near zero in the most critical cases
- Low dimensionality and noise lead to loss of separability
- t-SNE emerges as the most unstable method
- Algorithm differences diminish under extreme conditions

Observations and Features:	10000 observations and 5 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters		
Noise:	5	7	9	5	7	9	5	7	9
k-means	0,393	0,299	0,218	0,31	0,209	0,145	0,278	0,186	0,122
PCA + k-means	0,407	0,272	0,219	0,306	0,208	0,143	0,214	0,145	0,106
t-SNE + k-means	0,411	0,251	0,187	0,284	0,189	0,116	0,259	0,178	0,121
UMAP + k-means	0,5	0,304	0,186	0,34	0,226	0,154	0,301	0,204	0,124
Birch	0,33	0,287	0,184	0,208	0,132	0,098	0,293	0,169	0,095
PCA + Birch	0,245	0,255	0,181	0,292	0,186	0,107	0,28	0,204	0,127
t-SNE + Birch	0,329	0,078	0,144	0,23	0,191	0,133	0,212	0,169	0,133
UMAP + Birch	0,344	0,252	0,199	0,322	0,192	0,132	0,303	0,225	0,167
GMM	0,361	0,264	0,179	0,224	0,187	0,103	0,231	0,082	0,096
PCA + GMM	0,322	0,214	0,1	0,159	0,148	0,102	0,246	0,207	0,115
t-SNE + GMM	0,34	0,222	0,075	0,253	0,153	0,108	0,219	0,185	0,106
UMAP + GMM	0,501	0,27	0,115	0,301	0,193	0,159	0,299	0,196	0,127
PAM	0,438	0,349	0,229	0,314	0,2	0,144	0,306	0,188	0,116
PCA + PAM	0,446	0,341	0,217	0,312	0,204	0,134	0,215	0,153	0,112
t-SNE + PAM	0,417	0,251	0,186	0,284	0,192	0,118	0,266	0,18	0,113
UMAP + PAM	0,494	0,3	0,213	0,341	0,226	0,163	0,293	0,211	0,121
Spectral Clustering	0,464	0,31	0,221	0,321	0,204	0,137	0,302	0,194	0,129
PCA + Spectral Clustering	0,423	0,293	0,215	0,311	0,196	0,154	0,191	0,142	0,106
t-SNE + Spectral Clustering	0,336	0	0	0,334	0,167	0,145	0,267	0,195	0,129
UMAP + Spectral Clustering	0,554	0,313	0,198	0,334	0,232	0,138	0,275	0,196	0,129

RESULTS

Observations and Features:	2500 observations and 15 features									5000 observations and 10 features									10000 observations and 5 features								
Number of Clusters:	3 clusters			5 clusters			7 clusters			3 clusters			5 clusters			7 clusters			3 clusters			5 clusters			7 clusters		
Noise:	5	7	9	5	7	9	5	7	9	5	7	9	5	7	9	5	7	9	5	7	9	5	7	9	5	7	9
k-means	0,962	0,897	0,785	0,646	0,795	0,475	0,723	0,626	0,492	0,547	0,472	0,378	0,575	0,344	0,284	0,544	0,364	0,239	0,393	0,299	0,218	0,31	0,209	0,145	0,278	0,186	0,122
PCA + k-means	0,96	0,413	0,778	0,881	0,551	0,466	0,693	0,587	0,465	0,533	0,456	0,367	0,389	0,278	0,267	0,36	0,269	0,182	0,407	0,272	0,219	0,306	0,208	0,143	0,214	0,145	0,106
t-SNE + k-means	0,95	0,802	0,637	0,643	0,517	0,416	0,715	0,468	0,327	0,536	0,434	0,28	0,522	0,322	0,253	0,458	0,254	0,177	0,411	0,251	0,187	0,284	0,189	0,116	0,259	0,178	0,121
UMAP + k-means	0,961	0,892	0,792	0,949	0,837	0,466	0,73	0,591	0,372	0,549	0,446	0,355	0,622	0,342	0,334	0,473	0,307	0,198	0,5	0,304	0,186	0,34	0,226	0,154	0,301	0,204	0,124
Birch	0,925	0,843	0,676	0,92	0,52	0,561	0,794	0,537	0,4	0,618	0,433	0,244	0,401	0,291	0,261	0,447	0,239	0,214	0,33	0,287	0,184	0,208	0,132	0,098	0,293	0,169	0,095
PCA + Birch	0,963	0,883	0,731	0,637	0,53	0,397	0,658	0,543	0,268	0,397	0,306	0,285	0,352	0,276	0,24	0,331	0,242	0,169	0,245	0,255	0,181	0,292	0,186	0,107	0,28	0,204	0,127
t-SNE + Birch	0,94	0,822	0,67	0,658	0,578	0,418	0,706	0,393	0,315	0,52	0,304	0,222	0,403	0,361	0,157	0,359	0,207	0,179	0,329	0,078	0,144	0,23	0,191	0,133	0,212	0,169	0,133
UMAP + Birch	0,961	0,892	0,791	0,862	0,662	0,551	0,862	0,699	0,554	0,549	0,439	0,345	0,449	0,289	0,301	0,527	0,388	0,237	0,344	0,252	0,199	0,322	0,192	0,132	0,303	0,225	0,167
GMM	0,976	0,928	0,441	0,655	0,679	0,521	0,656	0,504	0,267	0,683	0,416	0,287	0,49	0,415	0,341	0,528	0,267	0,166	0,361	0,264	0,179	0,224	0,187	0,103	0,231	0,082	0,096
PCA + GMM	0,972	0,901	0,56	0,92	0,48	0,473	0,633	0,605	0,444	0,491	0,345	0,209	0,392	0,283	0,183	0,39	0,261	0,194	0,322	0,214	0,1	0,159	0,148	0,102	0,246	0,207	0,115
t-SNE + GMM	0,959	0,82	0,478	0,699	0,557	0,411	0,886	0,472	0,341	0,44	0,377	0,253	0,536	0,291	0,238	0,421	0,263	0,157	0,34	0,222	0,075	0,253	0,153	0,108	0,219	0,185	0,106
UMAP + GMM	0,961	0,899	0,792	0,953	0,835	0,493	0,735	0,629	0,293	0,494	0,371	0,251	0,511	0,305	0,231	0,47	0,259	0,237	0,501	0,27	0,115	0,301	0,193	0,159	0,299	0,196	0,127
PAM	0,887	0,427	0,4	0,899	0,473	0,24	0,647	0,421	0,222	0,396	0,393	0,298	0,389	0,305	0,225	0,479	0,289	0,177	0,438	0,349	0,229	0,314	0,2	0,144	0,306	0,188	0,116
PCA + PAM	0,961	0,889	0,813	0,613	0,533	0,446	0,688	0,591	0,459	0,544	0,457	0,368	0,328	0,261	0,261	0,345	0,233	0,17	0,446	0,341	0,217	0,312	0,204	0,134	0,215	0,153	0,112
t-SNE + PAM	0,95	0,808	0,662	0,643	0,516	0,424	0,717	0,475	0,321	0,537	0,433	0,291	0,399	0,324	0,252	0,45	0,235	0,18	0,417	0,251	0,186	0,284	0,192	0,118	0,266	0,18	0,113
UMAP + PAM	0,961	0,894	0,793	0,95	0,559	0,609	0,729	0,613	0,482	0,573	0,438	0,337	0,613	0,461	0,263	0,584	0,324	0,203	0,494	0,3	0,213	0,341	0,226	0,163	0,293	0,211	0,121
Spectral Clustering	0,971	0,915	0,794	0,957	0,853	0,685	0,949	0,825	0,674	0,56	0,446	0,354	0,61	0,449	0,317	0,647	0,444	0,247	0,464	0,31	0,221	0,321	0,204	0,137	0,302	0,194	0,129
PCA + Spectral Clustering	0,972	0,909	0,789	0,92	0,806	0,62	0,705	0,593	0,457	0,54	0,448	0,367	0,325	0,273	0,248	0,327	0,215	0,161	0,423	0,293	0,215	0,311	0,196	0,154	0,191	0,142	0,106
t-SNE + Spectral Clustering	0,961	0,763	0,582	0,938	0,475	0,33	0,946	0,443	0,287	0,547	0,371	0,33	0,348	0,279	0,217	0,454	0,221	0,152	0,336	0	0	0,334	0,167	0,145	0,267	0,195	0,129
UMAP + Spectral Clustering	0,445	0,898	0,796	0,954	0,836	0,565	0,703	0,623	0,444	0,558	0,397	0,323	0,632	0,438	0,278	0,473	0,442	0,233	0,554	0,313	0,198	0,334	0,232	0,138	0,275	0,196	0,129

Clustering performance measured by the Adjusted Rand Index (ARI)

HIGH DIMENSIONS

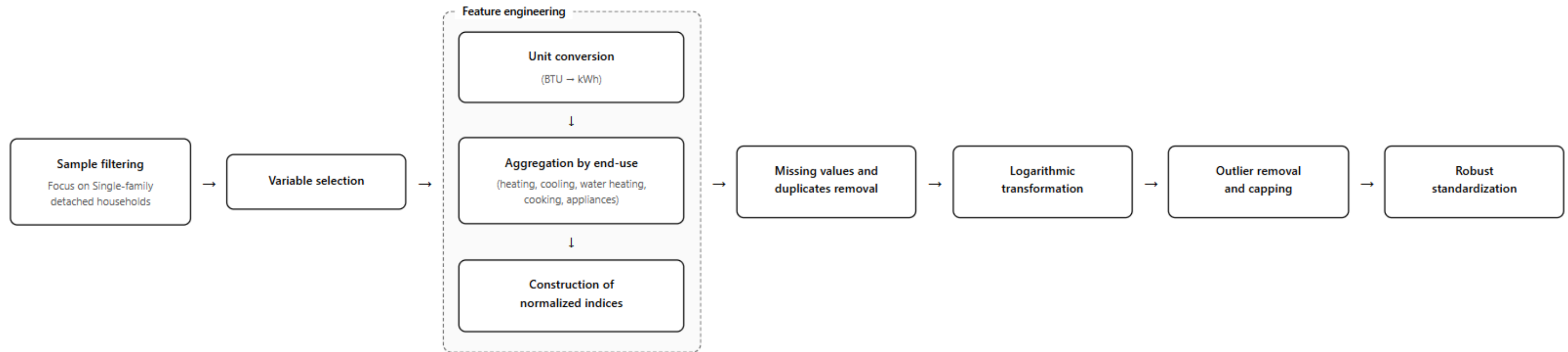
- In high-dimensional settings, the importance of dimensionality reduction is clearly evident.
- t-SNE and especially UMAP dramatically improve ARI and NMI for most methods.
- The benefit is particularly evident for k-means, GMM, and PAM.
- Spectral Clustering is less affected by preprocessing, consistent with its graph-based and compression nature.
- Low Silhouette scores reflect distance concentration effects in high-dimensional space (curse of dimensionality), reducing geometric separation despite high ARI and NMI

Algorithms	ARI	NMI	Silhouette Score
k-means	0,669	0,835	0,07
PCA + k-means	0,676	0,837	0,257
t-SNE + k-means	0,999	0,998	0,758
UMAP + k-means	0,999	0,998	0,919
Birch	0,811	0,908	0,093
PCA + Birch	0,762	0,906	0,293
t-SNE + Birch	0,766	0,905	0,638
UMAP + Birch	0,999	0,998	0,919
GMM	0,557	0,664	0,012
PCA + GMM	0,908	0,933	0,266
t-SNE + GMM	0,999	0,998	0,758
UMAP + GMM	0,999	0,998	0,919
PAM	0,528	0,636	0,059
PCA + PAM	0,745	0,868	0,295
t-SNE + PAM	0,76	0,904	0,65
UMAP + PAM	0,999	0,998	0,919
Spectral Clustering	1	1	0,101
PCA + Spectral Clustering	1	1	0,349
t-SNE + Spectral Clustering	0,488	0,757	0,209
UMAP + Spectral Clustering	0,318	0,647	0,217

Dataset composed of 2.500 observations, 100 variables, 7 clusters, correlation factor equal to 0.9 and noise equal to 9

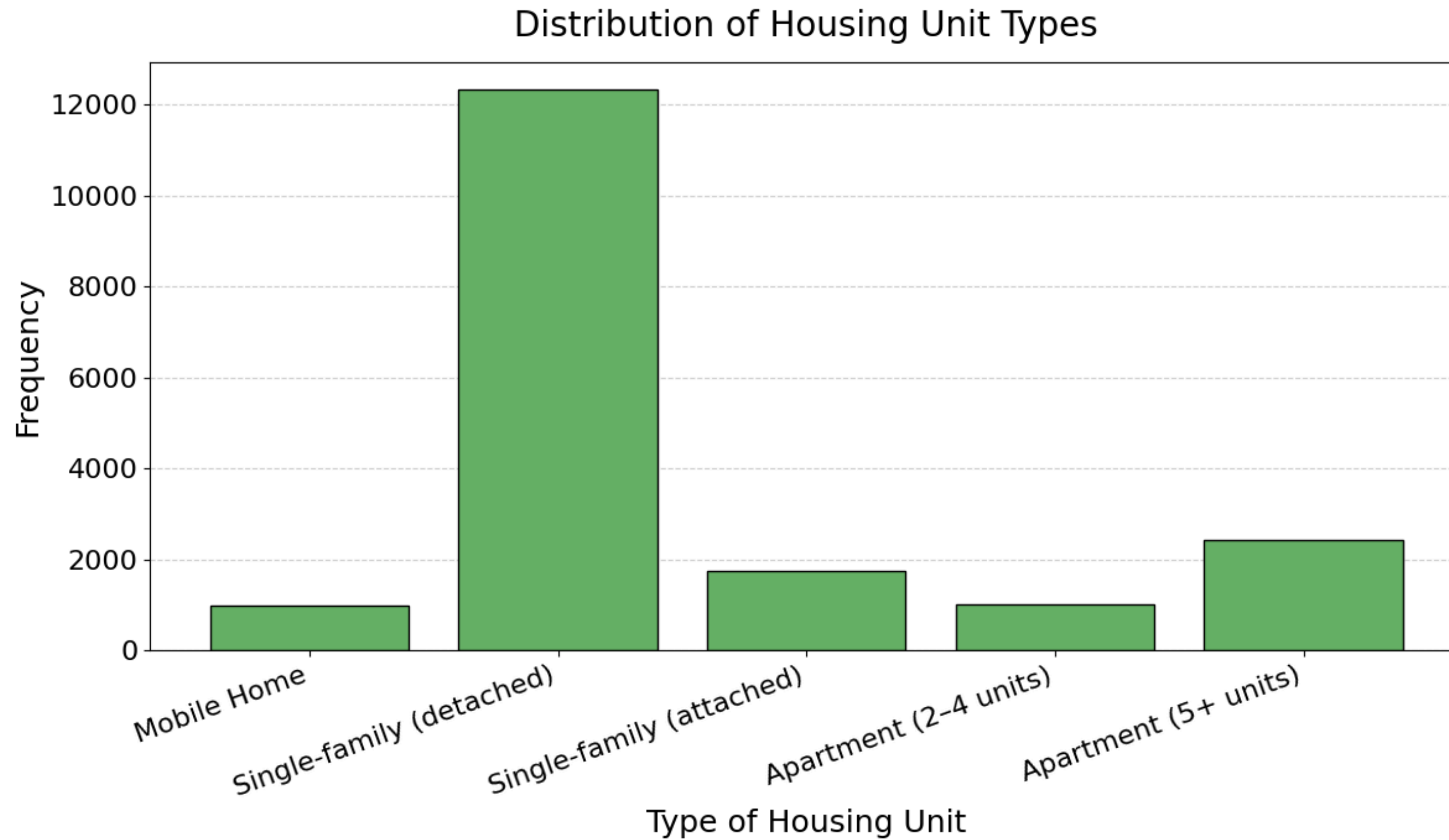
2 Validation on Residential Energy Consumption Data

Application on a Real Dataset

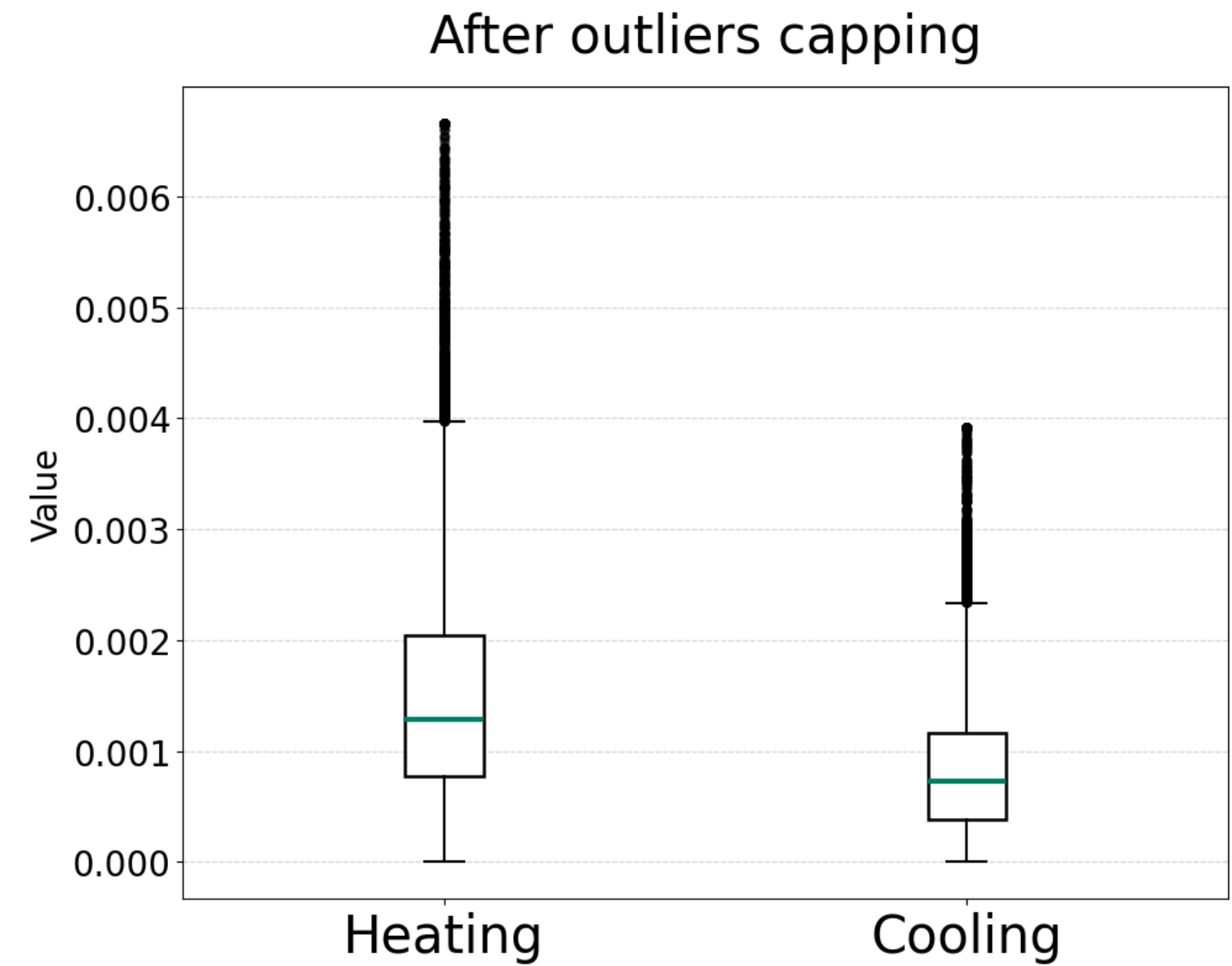
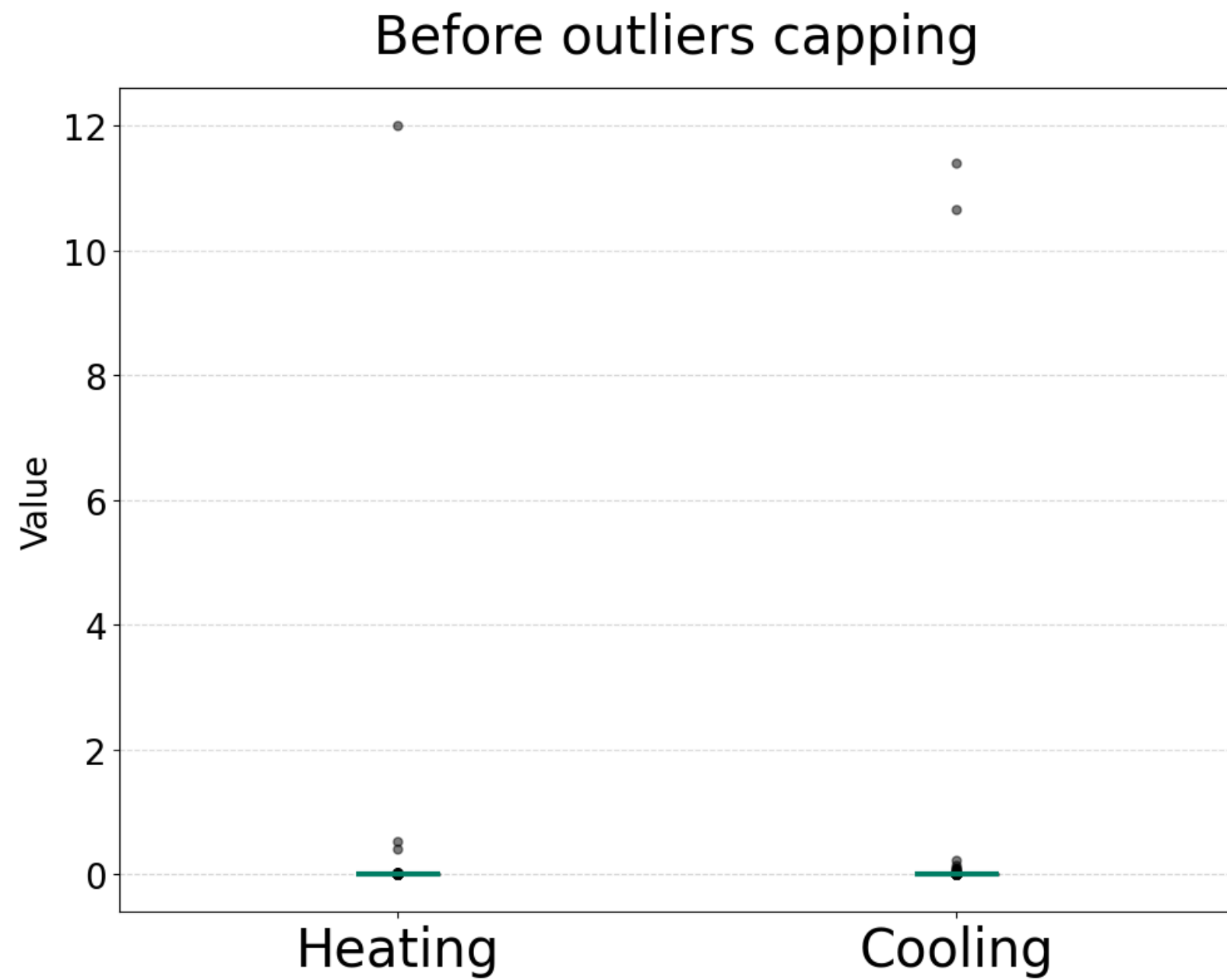


- **Dataset:** RECS 2020 with 18,500 households representing ~123.5M US homes, 799 variables on energy
- **Goal:** Analyze energy consumption patterns and identify clusters of households with recurring structures.
- **Validation:** Qualitative comparison of clustering solutions using internal validation metrics.
- **Pre-processing:** Multi-stage workflow to ensure consistency, quality, and comparability of variables before clustering.

WORKFLOW

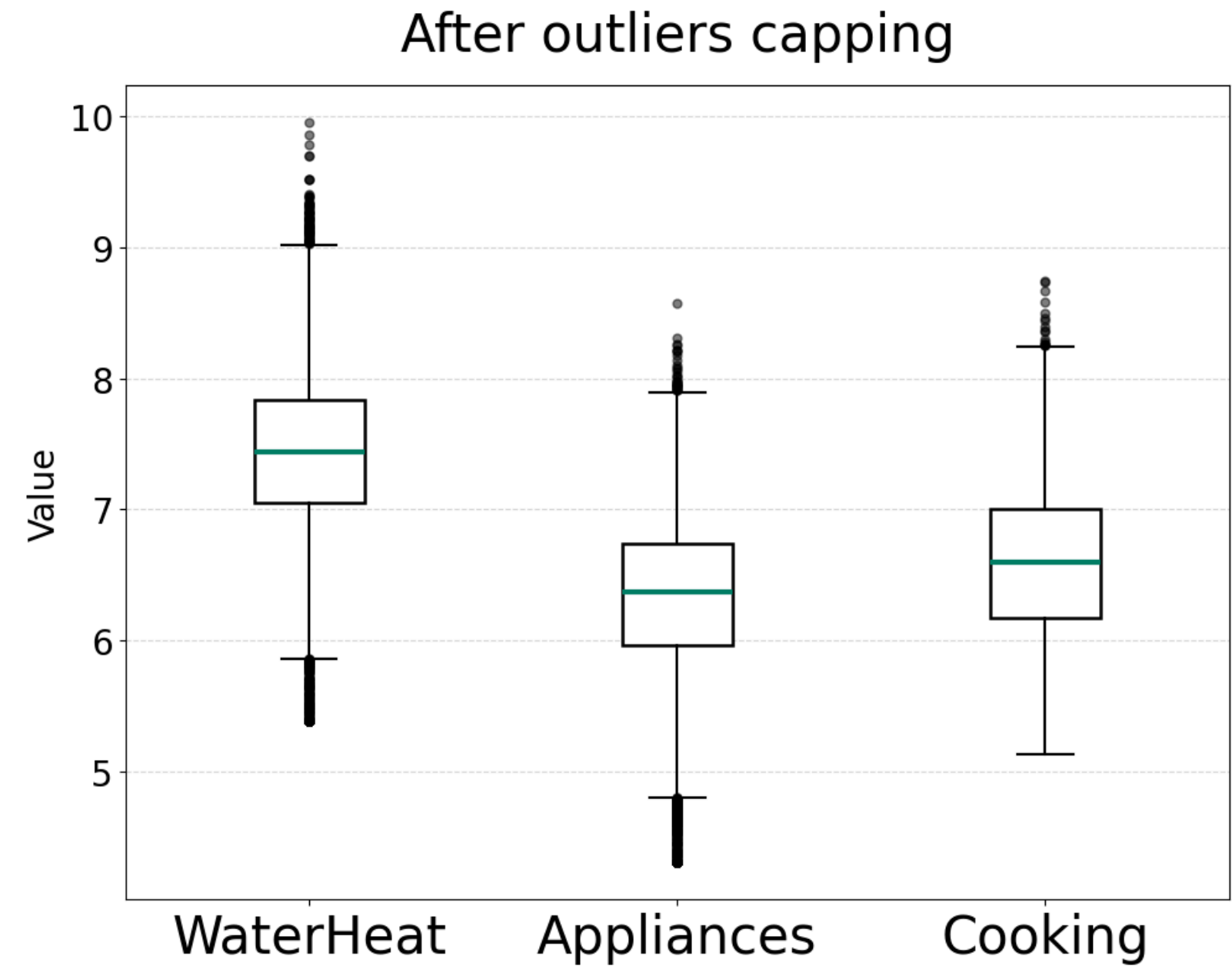
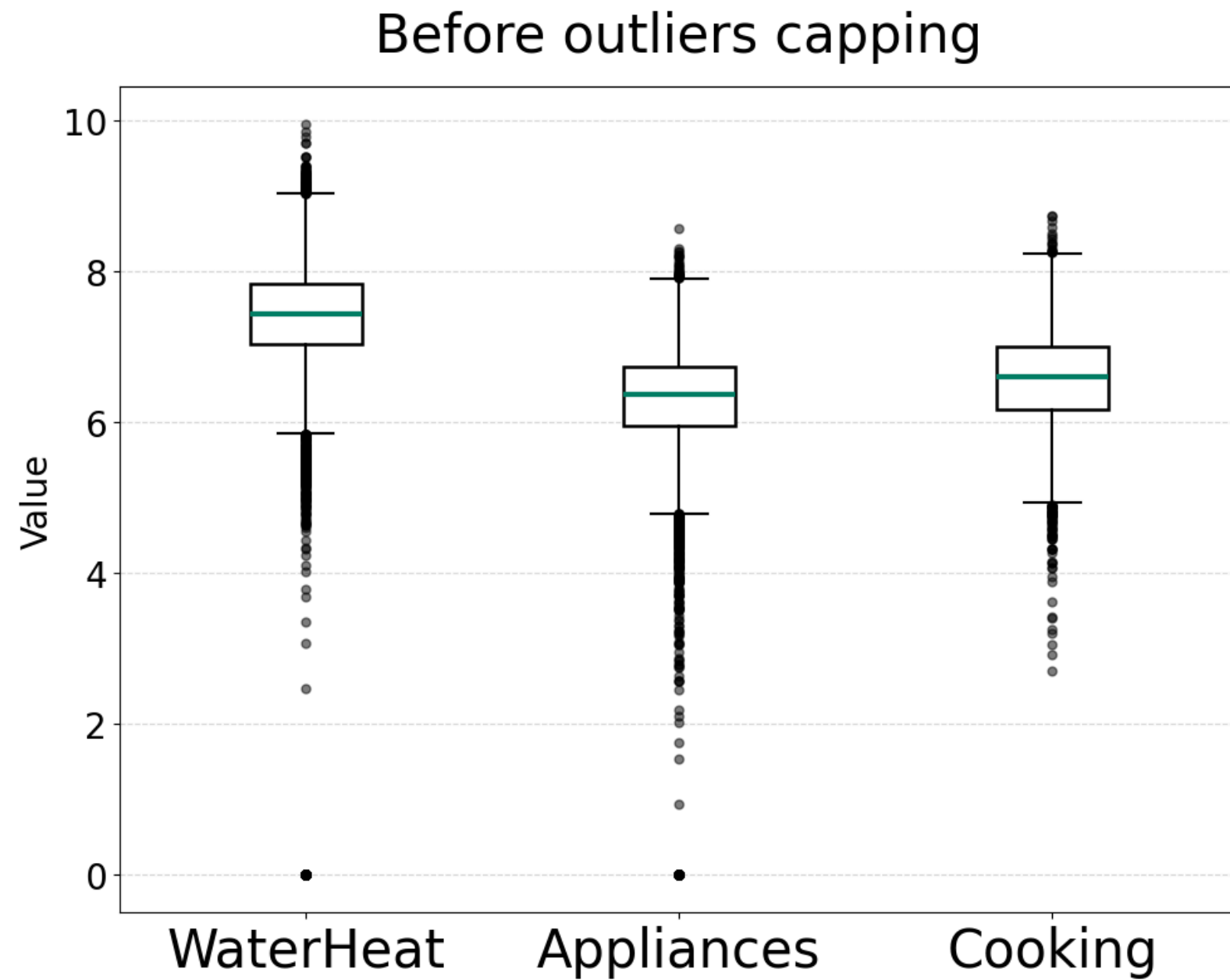


We restrict the analysis to households classified under the “Single-family (detached)” housing type.



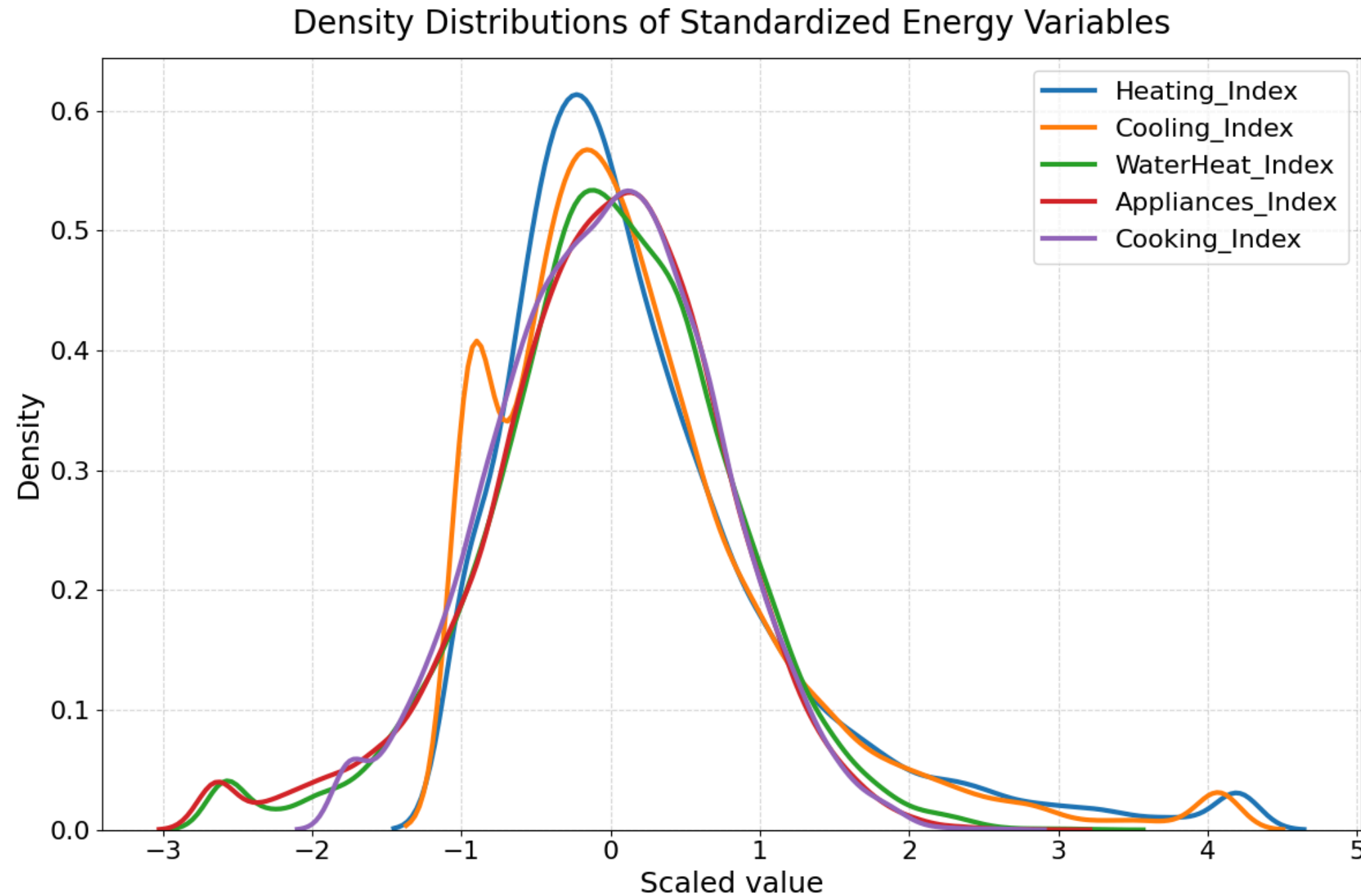
Outliers capping at the 99th percentile

OUTLIERS



Outliers removal and capping at the 1st percentile

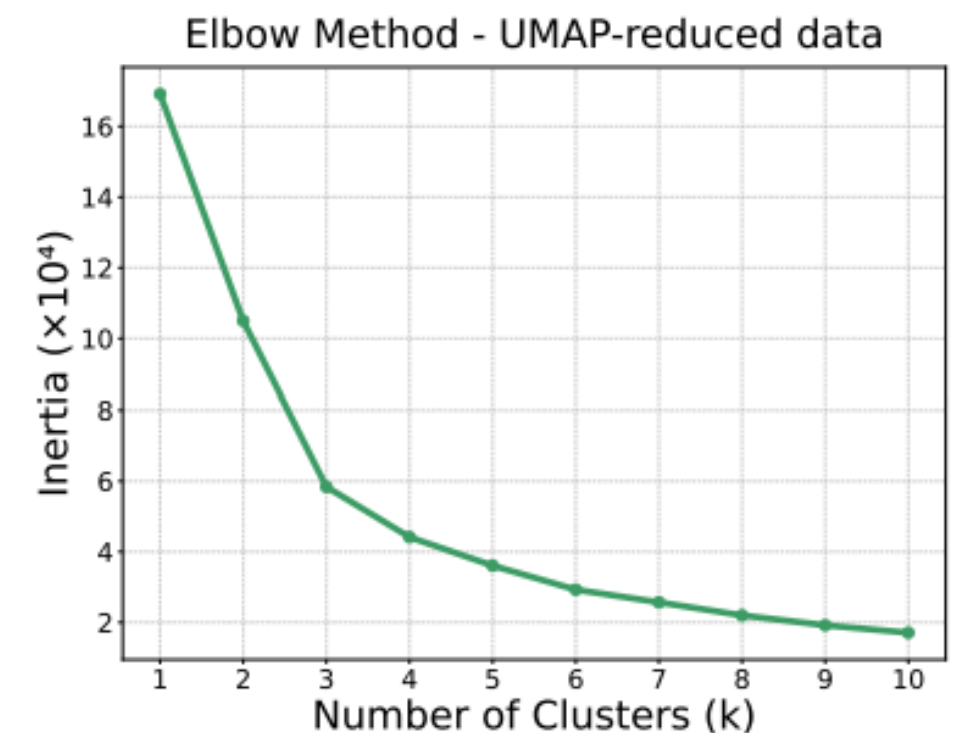
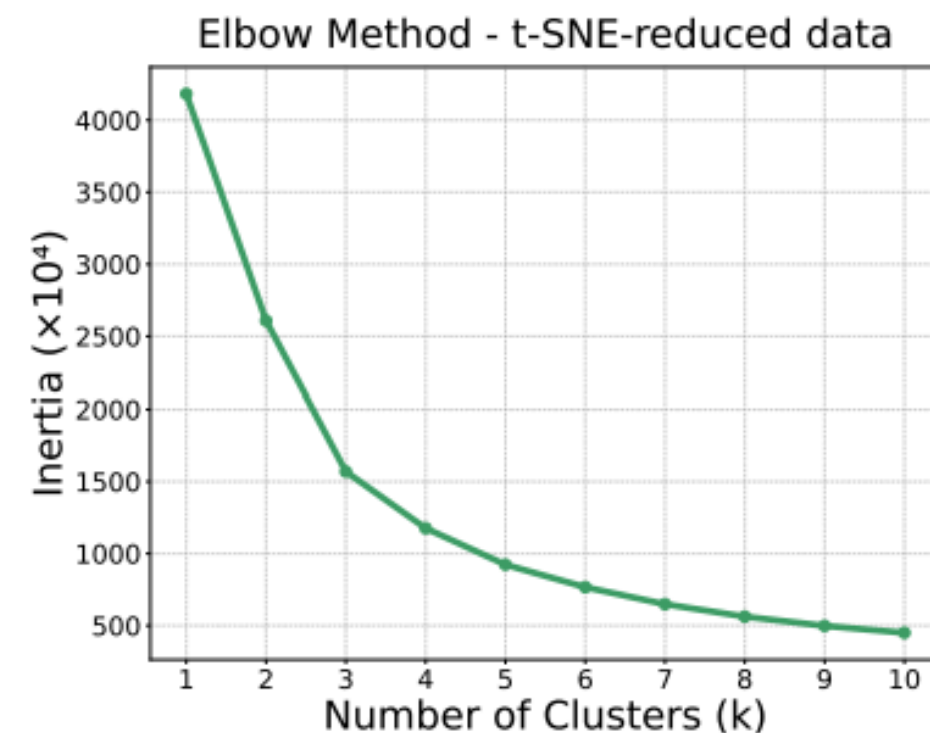
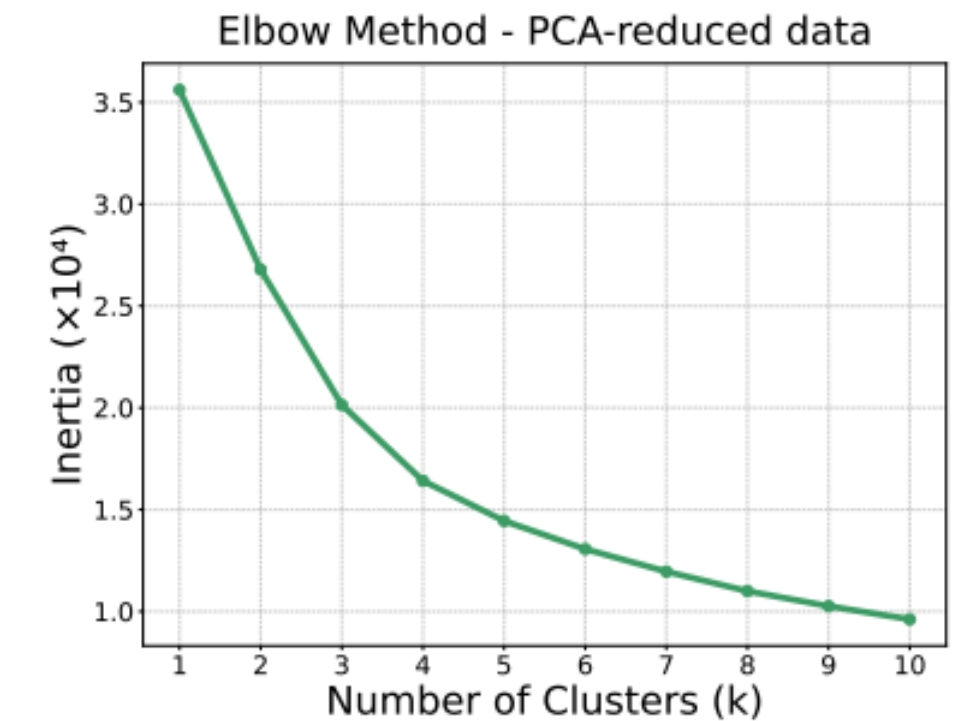
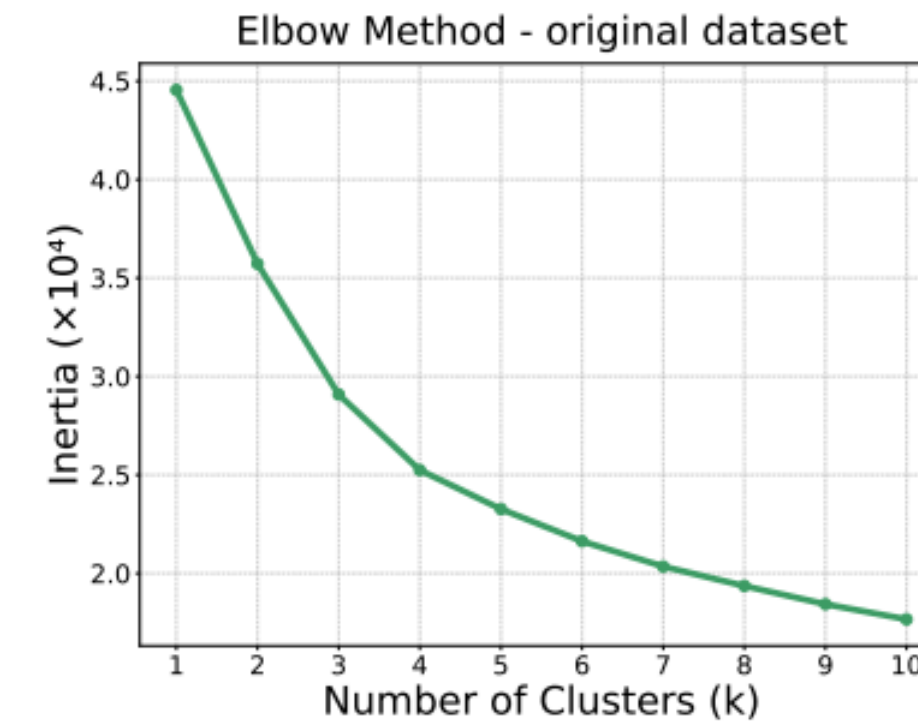
OUTLIERS



- Final distribution obtained after the dataset pre-processing stage.
- Slightly heavy tails are preserved, as they are informative for assessing the behavior and robustness of the different algorithms.

DISTRIBUTIONS

K	Original	PCA	t-SNE	UMAP
k=2	0.187	0.243	0.351	0.338
k=3	0.204	0.274	0.389	0.418
k=4	0.220	0.296	0.358	0.386
k=5	0.159	0.236	0.358	0.360
k=6	0.156	0.236	0.382	0.348



Elbow Method for k-means and Silhouette Scores

OPTIMAL K

Based on the comparative performance on internal evaluation metrics, we select the following configurations:

- PCA + k-means
- UMAP + PAM

Algorithms	Silhouette Score	Davies Bouldin	Calinski Harabasz
k-means	0,22	1,36	3116,279
PCA + k-means	0,296	1,097	4764,273
t-SNE + k-means	0,39	0,84	10175,179
UMAP + k-means	0,418	0,799	11601,247
Birch	0,195	1,495	1494,252
PCA + Birch	0,232	1,275	2170,844
t-SNE + Birch	0,329	0,986	7808,326
UMAP + Birch	0,354	0,895	8608,939
GMM	0,173	2,103	1535,094
PCA + GMM	0,211	1,462	2428,52
t-SNE + GMM	0,374	0,891	9374,913
UMAP + GMM	0,394	0,84	10368,128
PAM	0,21	1,393	3048,933
PCA + PAM	0,188	1,333	3882,545
t-SNE + PAM	0,39	0,841	10174,595
UMAP + PAM	0,418	0,798	11600,022
Spectral Clustering	0,221	1,32	2894,931
PCA + Spectral Clustering	0,242	1,227	4131,494
t-SNE + Spectral Clustering	-0,135	1,068	17,297
UMAP + Spectral Clustering	0,415	0,807	11383,167

ALGORITHM CHOICE

Label	Size	Heating	Cooling	Water Heating	Appliances	Cooking
0	1519	0.395	1.919	0.06	0.097	-0.173
1	1289	2.301	0.311	0.185	-0.058	0.063
2	5225	-0.029	-0.116	0.384	0.398	0.417
3	4195	-0.147	-0.142	-0.603	-0.685	-0.524

PCA + K-MEANS

Label	Size	Heating	Cooling	Water Heating	Appliances	Cooking
0	4428	-0.174	-0.194	0.397	0.349	0.459
1	4237	-0.126	-0.067	-0.608	-0.660	-0.571
2	3563	1.151	0.914	0.178	0.149	0.051

UMAP + PAM

Label	Age	Income	Employment	Education	Race
0	5.2	1335.5	Retired(47%) Full-time(40%)	College(29%) Bachelor(28%)	White(92%) Black(3%)
1	-6.5	4080.8	Full-time(57%) Retired(22%)	Bachelor(29%) College(28%)	White(83%) Asian(7%)
2	1.2	-6512.8	Full-time(46%) Retired(36%)	College(31%) Bachelor(24%)	White(86%) Black(9%)

UMAP + PAM

CONCLUSIONS

Goal:

Evaluate the behavior of multiple clustering algorithms in controlled and real-world settings.

Key drivers of performance:

Dimensionality, number of clusters, correlation, noise, and observation-to-variable ratio.

Methodological insight:

Clustering and dimensionality reduction should be designed as an integrated pipeline.

Dimensionality reduction:

UMAP is the most robust and consistent, especially in high dimensions.

Real-data evidence:

UMAP + PAM yields more interpretable and behaviorally meaningful clusters.

Future directions:

Non-Gaussian structures, systematic hyperparameter tuning, mixed-type data.

**THANK YOU
SO MUCH**
